

Indian Institute of Information Technology Allahabad

Support Vector Machine

Assignment - 5

From Maximum-Margin Hyperplanes to Comprehensive
Classification Evaluation

Objective

Understand maximum-margin classification, support vectors, kernels, SVM hyperparameters, and comprehensive classification evaluation using multiple metrics.

General Instructions

1. Use Python for all experiments.
2. NumPy, Pandas, Matplotlib, and scikit-learn may be used.
3. Use the same train/test split for comparable experiments.
4. Perform feature scaling using training data only.
5. Unless otherwise specified, use random state 42.
6. Explain every important numerical result.
7. All plots must have titles, axis labels, and readable legends.
8. Do not evaluate a classifier using accuracy alone.
9. Submit the notebook/source code and a short report containing tables, figures, observations, and conclusions.

Important

The SVM lecture emphasizes the maximum-margin hyperplane, support vectors, the mathematical optimization problem, kernels, and hinge loss. The assessment material emphasizes confusion-matrix-based metrics, ROC/AUC, and Precision–Recall analysis. Your assignment must connect both parts.

1 Part A: Dataset Understanding and Preprocessing

Use the **Breast Cancer Wisconsin Diagnostic dataset** available through scikit-learn.

A1 — Dataset Exploration

Load the dataset and report:

- number of samples and features;
- feature names;
- target classes;
- number and percentage of samples in each class.

Create a class-distribution plot.

A2 — Train/Test Split

Split the data into 80% training and 20% testing using stratification.

A3 — Standardization

Standardize each feature using

$$z = \frac{x - \mu}{\sigma}.$$

Calculate μ and σ from the training set only. Explain why scaling is important for SVM.

2 Part B: Maximum-Margin Hyperplane

Use the following two-dimensional dataset.

Class -1	Class +1
(1, 0.5)	(2, 0)
(1, 1)	(4, 0)
(1, -0.5)	(4.5, 1)
(-0.5, 0.5)	(4.5, 0.5)
(0.5, 0.5)	(5, -1)
	(5.5, 0)

B1

Plot the two classes.

B2

Train a linear SVM and obtain w, b . Write

$$w^T x + b = 0.$$

B3

For every point calculate

$$d(x, H) = \frac{|w^T x + b|}{\|w\|}.$$

Identify the closest points.

B4

Calculate the complete margin:

$$\gamma = \frac{2}{\|w\|}.$$

Identify and explain the support vectors.

Required figure: show both classes, the decision boundary, the two margin boundaries, and support vectors.

3 Part C: SVM Formulation and Hinge Loss

For $y_i \in \{-1, +1\}$, the hard-margin constraint is

$$y_i(w^T x_i + b) \geq 1.$$

The optimization problem is

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_i(w^T x_i + b) \geq 1.$$

C1

Explain why minimizing $\|w\|^2$ maximizes the margin.

C2

Implement the hinge loss

$$L_i = \max(0, 1 - y_i(w^T x_i + b))$$

using NumPy.

For the soft-margin formulation, implement

$$J(w, b) = \lambda \|w\|^2 + \sum_{i=1}^n \max(0, 1 - y_i(w^T x_i + b)).$$

C3

For selected points, calculate $y_i(w^T x_i + b)$ and determine whether each point satisfies the margin, lies inside the margin, or is misclassified.

4 Part D: Support Vector Analysis

Train a linear SVM on the standardized Breast Cancer dataset.

D1

Identify the support vectors and report

$$N_{SV}$$

and

$$SV\% = \frac{N_{SV}}{N_{\text{train}}} \times 100.$$

D2

Explain why points far from the decision boundary have less influence on the final boundary.

D3

Investigate

$$C \in \{0.01, 0.1, 1, 10, 100\}.$$

Report the number of support vectors, support-vector percentage, and test accuracy for each C .

5 Part E: Kernel SVM

Train SVM models using:

1. Linear kernel;
2. Polynomial kernel;
3. RBF kernel;
4. Sigmoid kernel.

Use the same training/testing data for all models.

For each model report:

- training accuracy;
- test accuracy;
- number of support vectors;

- training time, if measured.

Explain how the kernel trick permits nonlinear decision boundaries. Create:

Kernel	Train Acc.	Test Acc.	No. SVs	Time
Linear				
Polynomial				
RBF				
Sigmoid				

6 Part F: Hyperparameter Analysis

For the RBF kernel investigate

$$C \in \{0.01, 0.1, 1, 10, 100\}$$

and

$$\gamma \in \{0.001, 0.01, 0.1, 1\}.$$

Train one model for every (C, γ) pair. Record test accuracy and create a heatmap. Discuss:

- the best (C, γ) combination;
- possible underfitting;
- possible overfitting;
- the effect of increasing C ;
- the effect of increasing γ .

7 Part G: Comprehensive Classification Assessment

For the selected best model calculate the confusion matrix

$$\begin{pmatrix} TN & FP \\ FN & TP \end{pmatrix}.$$

G1 — Accuracy and Error Rate

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \quad Error = 1 - Accuracy.$$

G2 — Sensitivity and Specificity

$$Sensitivity = Recall = TPR = \frac{TP}{TP + FN}, \quad Specificity = TNR = \frac{TN}{TN + FP}.$$

G3 — False Positive and False Negative Rates

$$FPR = \frac{FP}{FP + TN}, \quad FNR = \frac{FN}{FN + TP}.$$

G4 — Precision and NPV

$$Precision = \frac{TP}{TP + FP}, \quad NPV = \frac{TN}{TN + FN}.$$

G5 — F1-score

$$F1 = \frac{2(Precision)(Recall)}{Precision + Recall}.$$

G6 — Matthews Correlation Coefficient

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}.$$

Explain the meaning of $MCC = 1$, $MCC = 0$, and $MCC = -1$.

G7 — Geometric Mean

$$GM = \sqrt{Sensitivity \times Specificity}.$$

Explain why GM is useful when both classes need to be considered.

Required table

Metric	Value
Accuracy	
Error Rate	
Sensitivity / Recall / TPR	
Specificity / TNR	
Precision	
F1-score	
FPR	
FNR	
NPV	
MCC	
Geometric Mean	

8 Part H: ROC Curve and AUC

For all four kernels, obtain continuous decision scores and plot

$$TPR \text{ against } FPR.$$

Calculate ROC-AUC for each kernel.

Kernel	ROC-AUC
Linear	
Polynomial	
RBF	
Sigmoid	

Answer whether the model with highest accuracy also has the highest ROC-AUC.

9 Part I: Precision–Recall Curve

Generate the Precision–Recall curve for each kernel:

$$Precision \text{ against } Recall.$$

Calculate PR-AUC.

Answer:

1. Which kernel has the highest PR-AUC?
2. Which has the highest ROC-AUC?
3. Are the rankings identical?
4. Does the highest-accuracy model also have the best PR performance?

10 Part J: Final Model Comparison

Create one comprehensive table.

Model	Acc.	Recall	Spec.	Prec.	F1	MCC	ROC-AUC	PR-AUC
Linear								
Polynomial								
RBF								
Sigmoid								

Include the final model's confusion matrix, selected hyperparameters, support-vector count, ROC curve, and PR curve.

11 Part K: Final Interpretation and Model Selection

Answer:

1. Which kernel has the highest accuracy?
2. Which has the highest recall?
3. Which has the highest specificity?
4. Which has the highest F1-score?
5. Which has the highest MCC?
6. Which has the highest ROC-AUC?
7. Which has the highest PR-AUC?
8. Which has the smallest FNR?
9. Does the best accuracy model remain best under other metrics?
10. Which model would you select for this medical classification problem, and why?

Selected SVM Model:

Justify the selection using at least five evaluation metrics.

Report and Submission Requirements

The submitted report must contain:

1. dataset dimensions and class distribution;
2. preprocessing procedure;
3. maximum-margin plot with support vectors;
4. calculated margin;
5. kernel comparison;
6. support-vector analysis;
7. C - γ heatmap;
8. confusion matrix;
9. all requested classification metrics;
10. ROC curves and ROC-AUC;
11. Precision-Recall curves and PR-AUC;
12. final comparison table;
13. final model-selection discussion.

Required Figures

1. Class-distribution plot.
2. Maximum-margin hyperplane with support vectors.
3. Loss/optimization plot where applicable.
4. C - γ heatmap.
5. Confusion matrix.
6. ROC curves.
7. ROC-AUC comparison.
8. Precision-Recall curves.
9. PR-AUC comparison.

Academic Integrity

All submitted code, plots, tables, interpretations, and conclusions must represent the student's own work. External code or sources used for implementation must be acknowledged.

End of Assignment